

Finding Unstructured References to Collective Agreements in French Legal Documents

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ABSTRACT

Collective agreements are documents that specify the terms and conditions of employment within a particular industry, such as the restaurant industry.

The task of detecting these agreements references in legal documents involves using text analysis and natural language processing techniques to identify and extract the agreements. The goal is to efficiently retrieve this information and link it to other relevant legal content, which can be useful for legal research and document analysis.

In this paper, an annotation scheme for this task is proposed and a dataset is built using it. An approach using dependency parsing for entity linking and embeddings for disambiguation is also proposed, achieving an F1-score of 0.82.

KEYWORDS

dataset creation, named entity recognition, entity linking, entity disambiguation, sentence embeddings, text processing for legal IR

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1 INTRODUCTION

France has over 650 collective agreements that specify the conditions of employment, each with a unique name. Extracting these names from unstructured and very long documents (1700 words on average), such as court decisions or laws, can be difficult due to their varying length, use of commas and acronyms. Moreover, collective agreements can be cited in multiple ways:

- base text (e.g. "convention collective des bureaux d'études techniques, des cabinets d'ingénieurs-conseils et des sociétés de conseils du 15 décembre 1987")
- attached text (attached to a collective agreement, see Figure 1 for more details) (e.g. "l'annexe n°2 du 12 juin 1999 de la convention",
- a specific article of a text (e.g. "article 3-2 de la convention").

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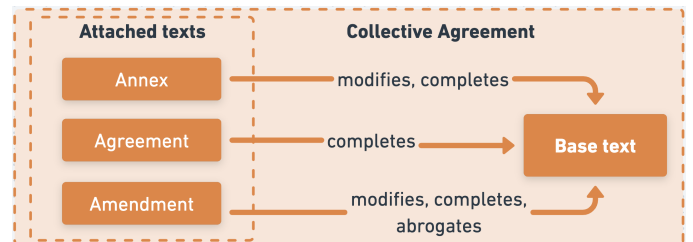


Figure 1: Collective agreement and attached texts

L' article 7 NUM_ART de l' avenant TEXT_ANNEX_CC n ° 31 NUM_LEG du
15 juin 2006 DATE_LEG relatif au nouveau statut du négociateur
immobilier TEXT_NAME annexé à la convention collective nationale
TEXT_CC de l' immobilier TEXT_NAME dispose à cet égard que les

Figure 2: Annotation scheme

Additionally, agreements may be mentioned implicitly (e.g. "article 3 of the applicable collective agreement"), and some agreements may have multiple articles with the same numbering (each chapter of the text has its own numbering).

2 ANNOTATION AND DATASET BUILDING

To the best of our knowledge, there is currently no annotated dataset available for this task. Therefore, we had to create one. The initial step involved using Named Entity Recognition. The annotation process can be challenging, as it involves dealing with implicit references, article numbers that may be distant from their corresponding text, and collective agreement names that can be lengthy or mentioned before or after they are first introduced.

Figure 2 shows an example of the annotation scheme we decided to use. Collective agreements are identified using the TEXT_CC tag, and their name using the TEXT_NAME tag.

Annotation was done manually but paragraphs were pre-annotated using a regular expression baseline. The data comes directly from the French institutional site Légifrance (or DILA) in charge of publishing legislative documents. We annotated a total of approximately 4050 paragraphs, which were taken from court decisions, commentaries, and law articles. However, we found that some collective agreements were rarely cited. In order to make our dataset as exhaustive as possible, we added additional paragraphs, by artificially replacing some collective agreement names with others from a list of all collective agreement names. For example, we might

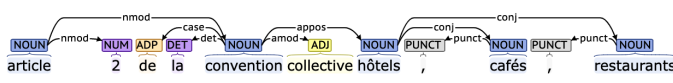


Figure 3: Dependency parsing

replace "convention collective de l'immobilier" with "convention collective de transport". We did this until we reached a total of 6000 paragraphs for our dataset.

Sometimes, when replacing a feminine noun with a masculine one, it can result in grammatically incorrect sentences (for example, "convention collective de la cinéma"). To our knowledge, there are various methods for augmenting data while preserving the linguistic coherence of generated sentences. [3] However, in our case, this was not a significant issue, as it could force the model to learn from context instead.

3 ENTITY DETECTION

First step is to detect all of the entities present in the text. Using regular expressions yields a poor F1-score of 0.1. This is because collective agreements name are lengthy, contain commas, and have multiple variations (such as "medical industry" and "medicine") etc... We decided to use a *bi-LSTM CRF* implementation, akin to the architecture used in the *Lample* paper [2], that we optimized using grid-search. This architecture works fine for this task because it handles capital letters (often the case for collective agreements names), and also uses a character based approach (handles acronyms).

4 ENTITY LINKING

Once the individual entities are extracted, it is necessary to link them together (for example, linking names to the corresponding collective agreements and connecting article numbers with the appropriate text). This is a dependency parsing task, for which we decided to use pre-existing French syntactic parsing modules. However, as none of these modules were specifically trained on legal data, we evaluated various tools. We ultimately chose the *stanza* implementation, along with manual rules to correct common errors and handle simple cases.

Figure 3 shows an example of a dependency parsing tree we get using stanza. We then use it to link the entities together with their closest candidate in the tree.

5 ENTITY MATCHING

With the entities linked, the next step is to identify the correct collective agreement that is being mentioned. Here are the details of the identification strategies in the most common cases:

CASE 1: if a collective agreement is explicitly mentioned (TEXT_CC with TEXT_NAME), we use the BM25 score between the TEXT_NAME with the set of available collective agreement names, applying stemming on the names. If the score is above a certain threshold, we return the correct collective agreement.

CASE 2: if a collective agreement is implicitly mentioned (TEXT_CC without TEXT_NAME), we retrieve the last mention to a collective agreement before this one, and use its TEXT_NAME as the

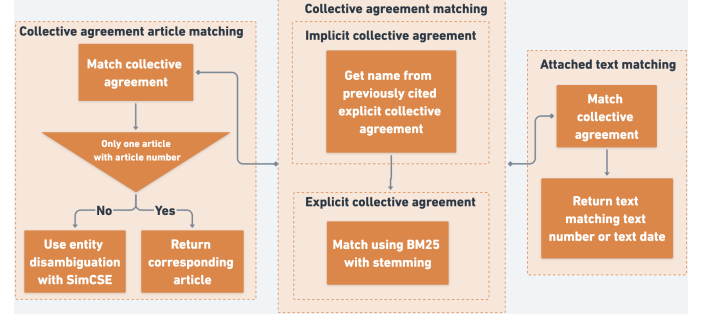


Figure 4: Entity matching

TEXT_NAME of the current entity. Then, we apply CASE 1

CASE 3: if a collective agreement is mentioned with an article (TEXT_CC with NUM_ART), we retrieve the right collective agreement via CASE 1 or CASE 2. If there is only one article number corresponding to the NUM_ART, we return this article. Otherwise we apply the disambiguation strategy detailed in the next section.

CASE 4: If an attached text article is mentioned (NUM_ART with TEXT_ANNEX_CC), we retrieve the correct collective agreement via CASE 1 or CASE 2. Then we retrieve the right attached text, using the NUM_LEG and DATE_LEG attached to the TEXT_ANNEX_CC. If there is only one article number corresponding to the NUM_ART, we return this article. Otherwise we apply the disambiguation strategy detailed in the next section.

Most common cases are detailed in Figure 4.

6 ENTITY DISAMBIGUATION

In the event that there are multiple articles with the same number in one collective agreement or attached text, we use an entity disambiguation scheme. The context in which the article is cited will likely be semantically similar to the article as they discuss similar themes. We first embed the whole paragraph where the collective agreement is cited, using a sentence embedding model *SimCSE* [1], trained on French legal documents. *SimCSE* is based on contrastive loss

$$\ell_i = -\log \frac{e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+) / \tau}}{\sum_{j=1}^N e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+) / \tau}},$$

We then also embed each candidate article using the same model, and return the one with the highest cosine similarity with the source paragraph (see cosine similarity distribution on Figure 5). If the score is below threshold (set to 0.3 to eliminate false positives), we output nothing. This allows us to handle 35% of the cases where there are multiple articles with the same number.

7 PERFORMANCE

For the Named Entity Recognition, we achieve a micro (weighted per tag occurrence) F1-score of 0.86. The recall (0.90) is slightly higher than the precision, but this is not really an issue, because

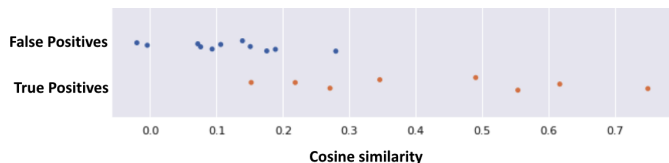


Figure 5: Cosine similarity scores distribution with SimCSE between false and true positives on a small dataset

in most cases, if an entity is wrongly detected, entity matching will not match it to anything. This performance is evaluated on a dataset without the artificially generated paragraphs using the replacement strategy explained in Section 2, in order to evaluate or approach on real data. On the overall extraction pipeline, we achieve a micro (finding all citations, counting duplicates) F1-score of 0.82, and a macro (finding all distinct citations) F1-score of 0.925.

8 CONCLUSION

We proposed an annotation scheme for this task, as well as a dataset boosting strategy to create artificial data. We trained an entity detection model, evaluated, and tested multiple entity linking strategies. Finally, we implemented an entity matching strategy based on different types of citations and achieved great results on a variety of legal documents.

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